How Does the Philippine Public Understand Dengue EWARS?: Evidence from an Urban Poor Community[1]

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Abstract

With the rollout of Ovicidal-Larvicidal Traps (OL traps) as a dengue early warning system (DEWARS) in various communities as a risk reduction and preparedness measure, the integration of effective risk communication is crucial, as it enables communities to understand the risks of dengue and strengthen their capacity to manage these risks. Using the Health Belief Model, statistical models were created to understand the real-world efficacy of OL traps through public understanding of its mechanism in the fight against dengue in the tropical Philippines. Data from sixty-eight participants from the urban poor community of Brgy. UP Campus, Quezon City, in the Philippines, was collected through a survey to peer at their understanding of the OL traps and the risks posed by mosquitoes and dengue. This investigation found that while the implemented DEWARS increases perceived susceptibility of residents, it does not translate to preventive knowledge-seeking. The findings also suggest that there is no significant difference between those households that received OL traps and those that did not in terms of dengue risk understanding. As such, there is a critical need to inform less-affluent communities of the science behind dengue and mosquito proliferation to mitigate the perpetuating risk of dengue.

Keywords: Dengue, risk communication, ovicidal-larvicidal traps, health belief model

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INTRODUCTION

The Philippines has a tropical maritime climate, experiencing perennial vulnerability to vector-borne diseases (Subido & Aniversario, 2022). The country's warm, humid environment and distinct rainy season provide an ideal, year-round breeding ground for the *Aedes aegypti* mosquito, making dengue fever an endemic and unavoidable public health threat (World Health Organization, 2024). While the risk of dengue can be found anywhere in the country, the incidence of an outbreak is not distributed equally, as population density and the level of poverty further increase the likelihood of dengue (Seposo, Valenzuela, & Apostol, 2023). The burden of dengue is overwhelmingly concentrated in densely populated urban centers, where a complex convergence of socioeconomic, environmental, and informational dilemmas exists (Araujo et al., 2020; Medina et al., 2025).

Dengue, as an urban health concern, is illustrated in Quezon City, the Philippines' most populous urban center, with a population of 2,960,048 and an average density of over 17,200 people per square kilometer (Philippine Statistics Authority, 2021). Within the city, this density is not uniform; districts containing large urban poor communities, such as District 2, have reported densities exceeding 31,000 people per square kilometer (Quezon City Government, 2020). Informal settlements often characterize these areas, and thus face structural challenges that amplify dengue risk. These risks include inconsistent solid waste collection, substandard housing, and a critical lack of reliable access to clean water, which compels residents to practice water storage in open containers, inadvertently creating a proliferation of mosquito breeding sites (Medina et al., 2025).

This urban public health concern is also compounded by systemic barriers to health information and access to health services. According to Atienza (2020), health literacy and access are inexplicably influenced by urban poverty. Urban poor communities often face financial constraints, exclusion from health decision-making processes, and difficulty in accessing timely medical services. Public health services in these areas are overwhelmed by its volume of clientele and are underfunded to meet patient needs and demands. These factors further widen the gap between urban poor individuals and families and their accessibility to critical health information and services, especially on dengue

prevention and symptoms. Dengue information may fail to reach them, or may not a perceived as a higher priority over more immediate concerns such as food and shelter.

The consequences of the complexity of dengue in urban poor areas are severe and quantifiable. According to the Quezon City Epidemiology and Surveillance Division (QCESD), there have been 9,243 dengue cases, of which 35 resulted in fatalities, from January 1 to October 20, 2025. This constitutes a 79.4% increase from the 5,151 cases during the same period in 2024 (Villanueva, 2025). Furthermore, the barangays that consistently reported the highest number of cases are Barangays Batasan Hills, Payatas, and Commonwealth (QCESD Weekly Epidemiological Surveillance Report [Week 41], 2025). These barangays are the same areas with the largest and most vulnerable urban poor populations. The current data alarmingly confirms that dengue transmission remains persistent among high-risk communities, despite existing knowledge and resources.

At present, the Dengue Early Warning and Response System (EWARS) is institutionalized by the Department of Health (DOH) as part of the Philippine Integrated Disease Surveillance and Response (PIDSR) framework (National Epidemiology Center - DOH, 2014). The system was designed to be a proactive and preventative measure that utilizes surveillance data to detect and predict outbreaks before they escalate (Hussain-Alkhateeb et al, 2018), triggering timely community-level interventions. Dengue surveillance begins with the Ovi-Larvicidal (OL) trap, a low-cost, community-based tool developed by the Department of Science and Technology (DOST) specifically to monitor Aedes mosquito density and provide a localized "early warning" (DOST, 2015).

Figure 1. OL Trap in Brgy. UP Campus

While EWARS exists as a national framework and OL traps have been deployed as a surveillance technology, a critical knowledge gap exists wherein it is fundamentally unknown how these two components are understood, integrated, and actioned at the grassroots level within the complex setting of an urban poor community. There is a presumed disconnect between the formation of entomological data from the OL traps and the practical use of this information within the dengue EWARS framework by local health workers and community members. It is unclear if stakeholders understand the OL trap's role beyond mosquito control and see it as a vital data source, which can be used to influence further action. Furthermore, it is unknown if this data actually triggers a timely "response" or if it is lost due to logistical, political, or social barriers found in resource-strained communities.

This paper, therefore, seeks to investigate the functional understanding and practical application of OL trap surveillance data within the dengue EWARS framework in the urban poor communities of Quezon City. Specifically, it sought to answer the following research questions:

1. Does receiving an OL trap affect participants' perception of dengue risk?
2. What factors are significant predictors of participants' accurate understanding of dengue risk?
3. How do participants view OL traps as an early warning system against dengue?

Using a mixed-method approach, this paper will attempt to explore the answers to these questions and forward a set of recommendations that may be adopted by researchers and policymakers alike in managing risks of dengue in the Philippines in a resource-limited context.

LITERATURE REVIEW

Dengue Situation in the Philippines

Dengue remains a major public health threat in the Philippines, which consistently reports some of the highest cases in Southeast Asia (Undurraga et al., 2013). Although outbreaks peak during the wet season from June to February (Undurraga et al., 2017), dengue transmission is increasingly recognized as a complex, year-round issue shaped by social and environmental conditions (Cordero, 2024). As of August 13, 2025, the Department of Health (DOH) recorded 136,161 dengue cases, a 33% increase from the 102,374 cases in the same period last year, although deaths declined from 401 to 364 (De Vera-Ruiz, 2024). In February 2025, Quezon City declared a dengue outbreak after reporting 10 deaths and over 1,700 cases, with children aged 1–10 years accounting for the highest burden (4,148 cases) (QCESD, 2025).

Early Warning and Response Systems (EWARS) in Dengue

In the global context, Early Warning and Response Systems (EWARS) emerged from a necessity in preventing local disease outbreaks from escalating into humanitarian disasters. The World Health Organization (WHO) institutionalized EWARS as an active public health intervention to monitor and predict local dengue outbreaks, given that a warming climate leads to increasing temperatures, higher rainfall, and humidity. This thereby increases the risk of spread of the dengue epidemic (WHO, 2025). At present, EWARS is a framework adopted by various nations, including the Philippines.

In the Philippine context, the lead public institution in the implementation of Dengue EWARS is the DOH alongside the local government units (LGUs), as part of the Philippine Integrated Disease Surveillance and Response (PIDSR) framework (National Epidemiology Center - DOH, 2014). The PIDSR framework mandates LGUs in reporting notifiable diseases to trigger specific protocols, such as the "4S-Strategy" for dengue: search and destroy mosquito breeding sites, self-protection among communities and families, seek early health consultation on dengue symptoms, and support local health initiatives, such as fogging and spraying areas where there have been dengue cases (Oliverio, 2024). While

the PIDSR framework provides the structure for response, there remain challenges in the *early warning phase*. Currently, the DOH implemented the Epidemic Prone Disease Surveillance Information System (EPDSIS), which is an online centralized database of information on reported cases, outbreaks, and other critical data for timely dengue monitoring (Mendoza et al., 2023). However, this system continues to face numerous challenges, such as the gap between the capacity of local health workers to ensure the accuracy of data and the timeliness of reporting. Additionally, there are duplications of reporting systems that are being implemented simultaneously, which leads to inefficiencies in data collection and analysis (Mendoza et al., 2023).

OL traps are an effective community-based tool against Aedes aegypti, with recent innovations integrating IoT and Convolutional Neural Networks to automate egg detection and data reporting through systems like C-DEWS (Aguinaldo & Sicam, 2025). Such technology allows monitoring of mosquito activity, environmental factors, and egg counts, enhancing early dengue surveillance. Moreover, studies show strong community acceptance, with users recognizing their effectiveness and expressing willingness to adopt them, although proper guidance on the disposal of larvicidal solutions remains essential (Ernesto, 2014). These findings indicate that OL traps, particularly when combined with digital monitoring, offer a practical and widely supported strategy for dengue control.

Public Understanding and Risk Communication in DRM

Risk communication ensures that experts and public officials can effectively convey risk-related information to the public in order to mitigate its impacts on health, and economic and social well-being (World Health Organization, n.d.). During public health emergencies, risk communication is crucial in informing people about existing health risks they face and strengthening their capacity to manage such risks by adopting appropriate measures to protect themselves (World Health Organization, 2017).

For risk communication to be effective, data providers must recognize their audiences and the factors that shape their perception and understanding of risks by considering language barriers, literacy levels, and the public's familiarity with technical terms (Jensen, 2024). A study by Akakpo and Neuerer (2024) found that individuals' acceptance of and adherence to health campaigns are tied to their health

literacy, or their ability to understand health-related information, whether personal or disseminated through public health warnings. Failure to consider relevant factors may render any risk reduction initiatives ineffective, thereby retaining the vulnerability of communities despite the presence of regular health campaigns.

As an integral component in crisis response and management of local government units (LGUs), risk communication should not simply take place in a centralized, top-down manner. Apart from ensuring that risk communication is considered, it should also be strategically delivered and rightfully done so by making public understanding of primary importance (Flores & Asuncion, 2020). Furthermore, community engagement should also be integrated with risk communication initiatives of LGUs in order to contextualize the present health issues among the constituents and enable behavior change, particularly improved health-seeking behavior (Dayapera et al., 2024).

Factors Influencing Public Understanding of EWARS and Dengue Prevention

A study in Malaysia found that low-income families practiced stronger dengue prevention due to their higher exposure to mosquito breeding sites in poorly maintained environments. Unemployed individuals were also more likely to engage in prevention than employed individuals, who were often preoccupied with work. Those with prior dengue experience showed greater knowledge and awareness (Wong, 2015).

Higher perceived susceptibility was linked to stronger dengue prevention practices, while low risk perception reduced motivation to act. Meanwhile, although residents recognized the link between climate change and dengue, many were uncertain about how outbreaks could be predicted. Media platforms like television and social media play key roles in spreading EWARS information, but must be complemented by interactive educational materials for deeper understanding (Zaki et al., 2019).

The Health Belief Model in Dengue EWARS of Brgy. UP Campus

This study is anchored on the Health Belief Model (HBM) developed by Rosenstock (1974), which explains how individuals' beliefs and perceptions influence their engagement in health-related behaviors. The model consists of six components: perceived susceptibility, or one's belief about the likelihood of acquiring a disease; perceived severity, or the belief about the seriousness and consequences of the illness; perceived benefits, or the belief in the effectiveness of the recommended preventive action; perceived barriers, or the perceived obstacles that hinder the behavior; cues to action, which are the triggers that prompt an individual to act; and self-efficacy, or the confidence in one's ability to perform the desired behavior.

Applied to this study, the HBM provides a framework to understand how residents perceive and respond to the Dengue Early Warning and Response Systems (DEWARS) through OL Traps. Residents' perceptions of risk (susceptibility and severity), the usefulness of EWARS (benefits), and possible challenges (barriers) affect how they understand and participate in dengue prevention efforts. Meanwhile, educational campaigns and communication materials serve as cues to action that enhance self-efficacy and support informed community participation. Thus, the HBM explains how individual beliefs and information cues collectively shape the public understanding of dengue early warning systems in urban poor communities of Quezon City.

CONCEPTUAL FRAMEWORK

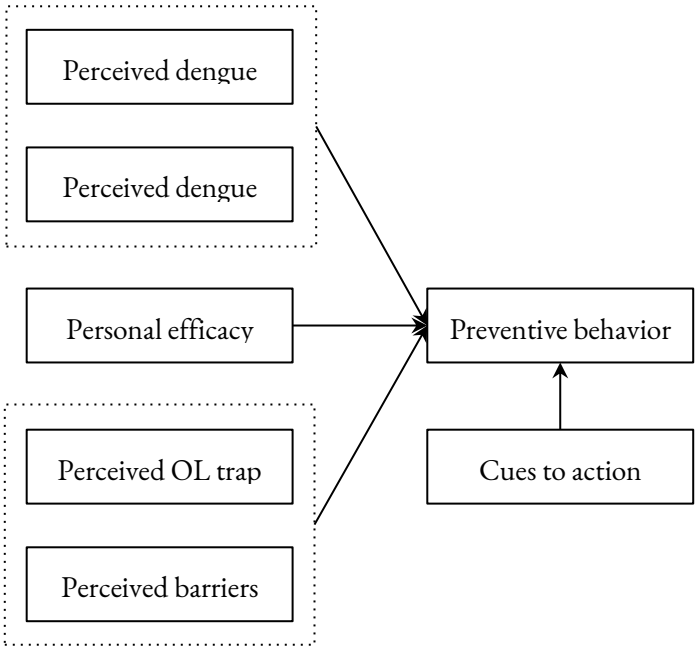
Guided by the HBM, this study assumes that residents' understanding of DEWARS through OL Traps is shaped by their perceptions of risk, benefits, and barriers, as well as by the presence of effective communication cues.

The OL traps, as an independent variable, represent the intervention under DEWARS that allows residents to recognize the presence of dengue vectors in their area. Their visibility can heighten residents' perceived susceptibility and severity of dengue, influencing awareness and concern about prevention. The materials for understanding DEWARS (mediating variable), such as orientations,

posters, or information drives, serve as cues to action that mediate the relationship between OL traps and public understanding. These materials help clarify the function of the traps, strengthen self-efficacy, and reduce perceived barriers to participation in dengue prevention. The public understanding of DEWARS (dependent variable) reflects how residents interpret and respond to EWARS initiatives. Under the HBM lens, this understanding is influenced by how they perceive dengue risk and by how effectively information materials guide their awareness and engagement.

Overall, the framework posits that the presence of OL traps enhances public understanding of dengue early warning systems when supported by clear and accessible educational and communication materials that motivate community action.

Figure 2. HBM as a Conceptual Framework for DEWARS

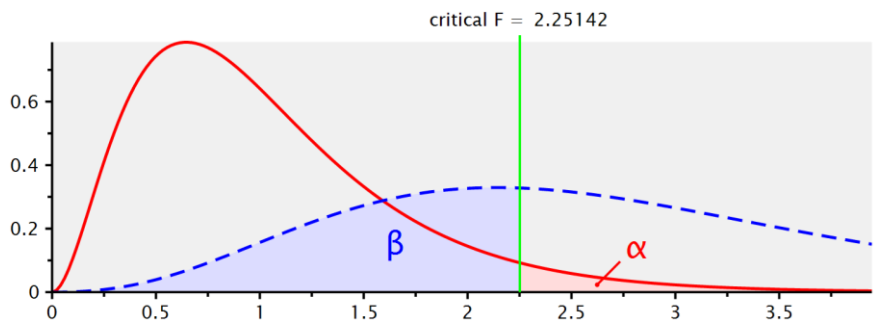


METHODOLOGY

A multivariate linear regression was created using the Health Belief Model. Five predictors were used in the statistical model: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, and self-efficacy. The outcome variable is the public's cue to action, aligned with the dengue early warning systems rolled out in their community.

Data was collected from 68 participants in Brgy. UP Campus, an urban community in Quezon City, Philippines. The was calculated using an effect size (f^2) = 0.15, α error prob = 0.05, six predictors, and power = 0.60.

Figure 3. Statistical Power and Error (Type I and Type II) in the F-Distribution



Each participant was provided with a survey questionnaire that could be accessed digitally or with pen and paper. The questionnaire was divided into three sections: (1) personal information, (2) perceptions concerning dengue EWARS, and (3) four open-ended questions regarding dengue EWARS. The questionnaire was encoded in Filipino for readability and access.

The data was manually scrubbed and then loaded into R for statistical analysis. For each predictor, the standard p-value of 0.05 was set to determine the statistical significance of its effect on public perception. The direction of each predictor was also observed, as well as the adjusted R^2 for precise model fitting. The qualitative data harnessed from Section 3, meanwhile, were analyzed through inductive reflexive thematic analysis (RTA). The answers were first coded into several categories, then grouped into themes to support the findings of the statistical analysis.

RESULTS AND DISCUSSION

This section will discuss the empirical findings of the fieldwork conducted by the researchers during the month of November 2025 at Brgy. UP Campus, Quezon City.

Demographic Characteristics of Participants

The participants consisted of 68 individuals drawn from seven urban poor communities in Brgy. UP Campus, Quezon City. These communities received OL traps that are monitored weekly. The participants were evenly divided between households that received OL traps (treatment group, n = 34) and those that did not (control group, n = 34). All participants from the control group were randomly surveyed, while those respondents in the treatment group were selected based on data provided by Brgy. UP Campus is identifying households that received OL traps. The majority of the participants in both groups were female, with 27 respondents each, thus comprising nearly 80% of the total sample. In terms of age distribution, the sample was slightly skewed toward the middle-aged population, particularly those aged 45-59 (n = 23), while older adults aged 60 and above had the most limited of respondents (n = 12).

Table 1. Age and Sex Distribution of Participants per OL Distribution Group

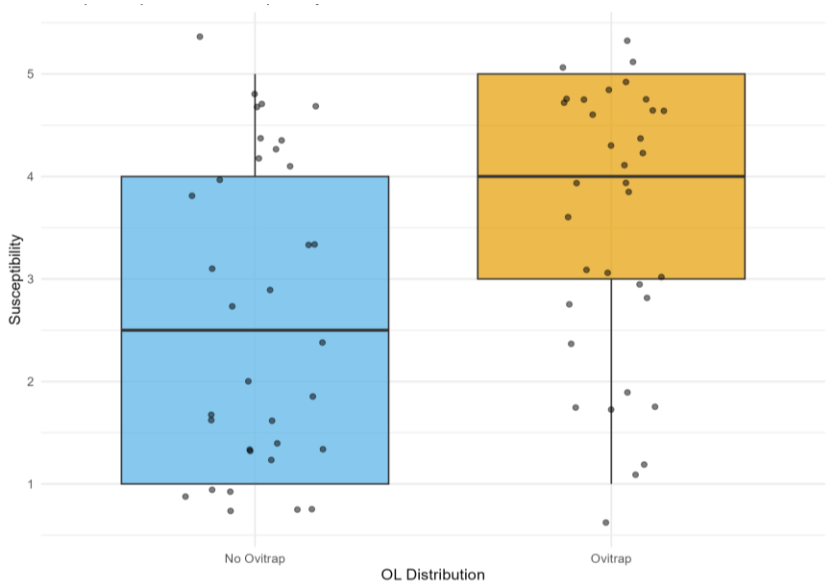
	Age Group				Total	% n
	18-29	30-44	45-59	60+		
Treatment Group	10	11	8	5	34	50.00%
Male	0	2	2	3	7	10.29%
Female	10	9	6	2	27	39.71%
Control Group	4	8	15	7	34	50.00%
Male	1	2	2	2	7	10.29%
Female	3	6	13	5	27	39.71%

OL Trap Rollout's Impact on Dengue Risk Perceptions

Using an independent samples t-test, six variables from the HBM were used to measure the participants' motivational beliefs on dengue and dengue prevention. This analysis compared the results of those who received the OL traps (treatment) and those who did not (control).

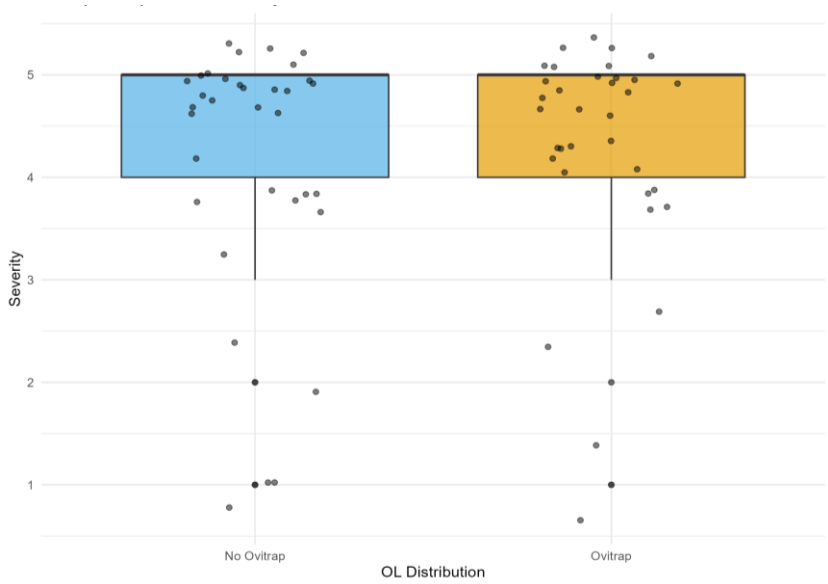
In terms of perception of susceptibility, the OL trap intervention was highly successful in altering perceived dengue risk. The mean susceptibility score for the treatment group ($\bar{x}$ = 3.62) was significantly higher than the control group ($\bar{x}$ = 2.68, p = 0.008). The analysis also showed that the intervention has an effect of 0.67, suggesting that the physical presence of the OL trap serves as a powerful and tangible motivator for preventive action.

Figure 4. Comparison of Perceived Susceptibility by OL Distribution Group



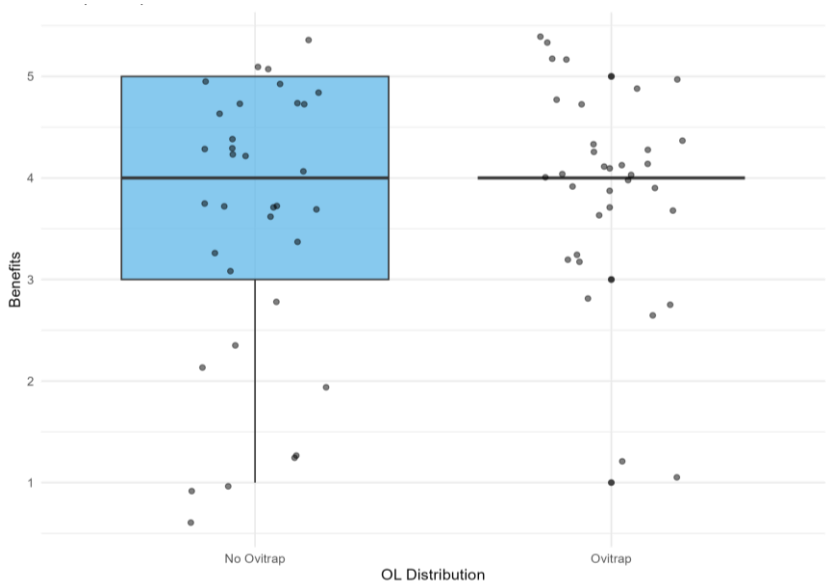
The program had no discernible effect on perceived severity in both groups (d = 0.07). Both groups scored highly and almost identically, evidence that there is a ceiling effect on dengue severity ($\bar{x}_{treatment}$ = 4.29, $\bar{x}_{control}$ = 4.21, p = 0.762). This would suggest that dengue is already perceived as a serious health threat within the community. The OL trap did not need to, nor was it able to, intensify this belief.

Figure 5. Comparison of Perceived Severity by OL Distribution Group



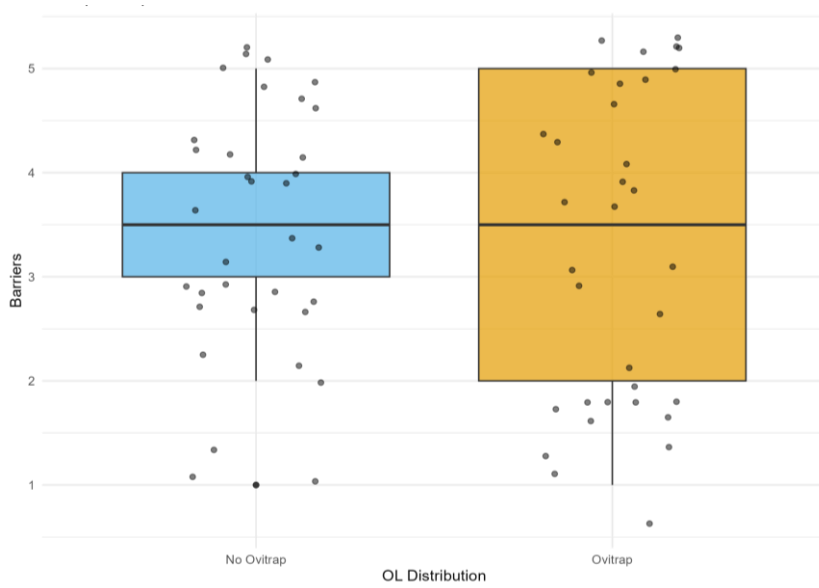
Similarly, there is also a small difference in the perceived benefits between the two groups ($\bar{x}_{treatment} = 3.88$, $\bar{x}_{control} = 3.56$), which was not statistically significant ($p = 0.273$). While there is a slight positive trend ($d = 0.27$), receiving the physical tool did not strongly cement the belief that the OL trap was an effective or crucial measure. This finding will be further discussed in the latter section of this paper.

Figure 6. Comparison of Perceived Benefits by OL Distribution Group



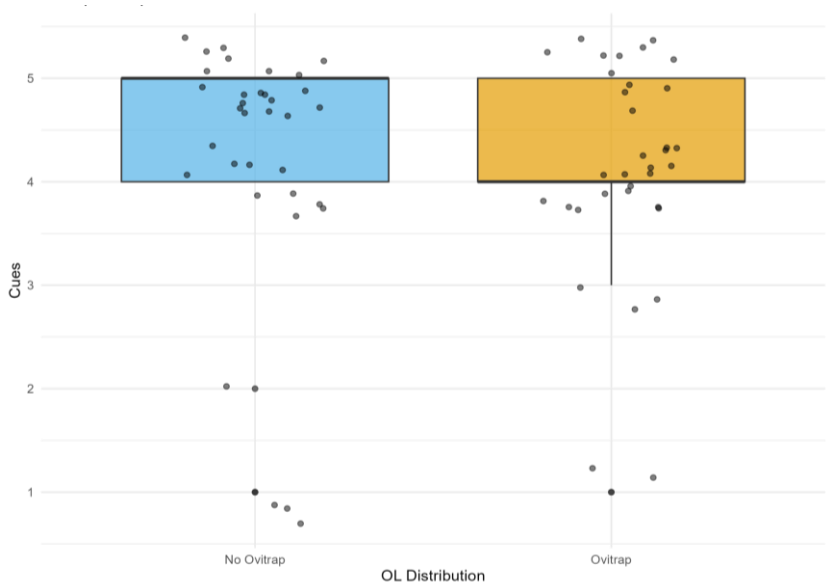
In terms of perceived barriers, there is a small decrease in the effect ($d = 0.27$) in the treatment group ($\bar{x} = 3.29$) compared to the control ($\bar{x} = 3.47$). However, this effect is negligible and non-significant ($p = 0.588$). This would indicate that receiving the OL trap did not meaningfully reduce participants' perception of barriers associated with dengue prevention and mosquito proliferation.

Figure 7. Comparison of Perceived Barriers by OL Distribution Group



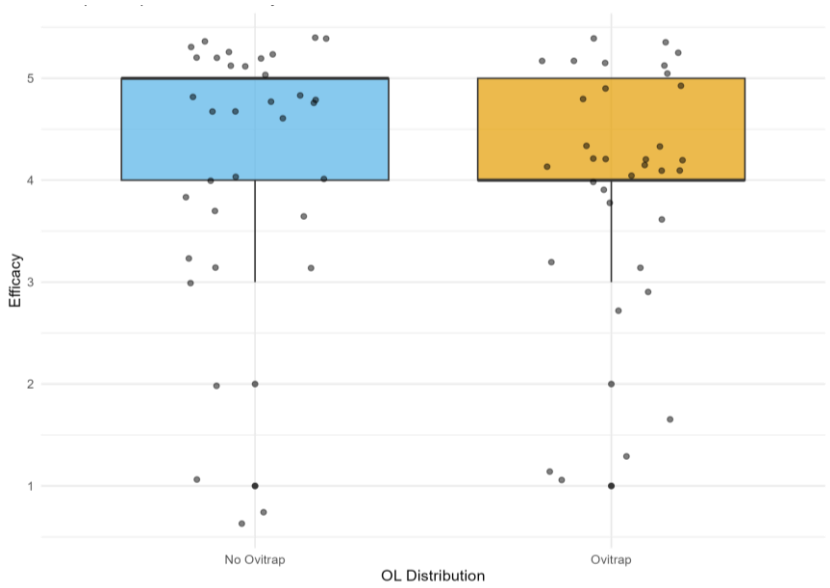
The rollout also did not significantly impact participants' cues to action, and scores across groups are similar ($\bar{x}_{treatment} = 4.09$, $\bar{x}_{control} = 4.26$, $p = 0.515$). This would suggest that the participants' engagement in preventive behavior is already high and is not triggered by the OL trap. A reflexive thematic analysis will supplement this finding in the next section.

Figure 8. Comparison of Cues to Action by OL Distribution Group



Scores in self-efficacy were also not significantly different between groups ($\bar{x}_{treatment} = 3.88$, $\bar{x}_{control} = 4.15$, $p = 0.378$). These critical findings show that the provision of the trap did not translate into greater confidence among participants in their ability to control dengue in their homes successfully. This would arguably mean that the OL traps were seen as a more passive tool rather than an integral part of the general management of dengue risk.

Figure 9. Comparison of Self-Efficacy by OL Distribution Group

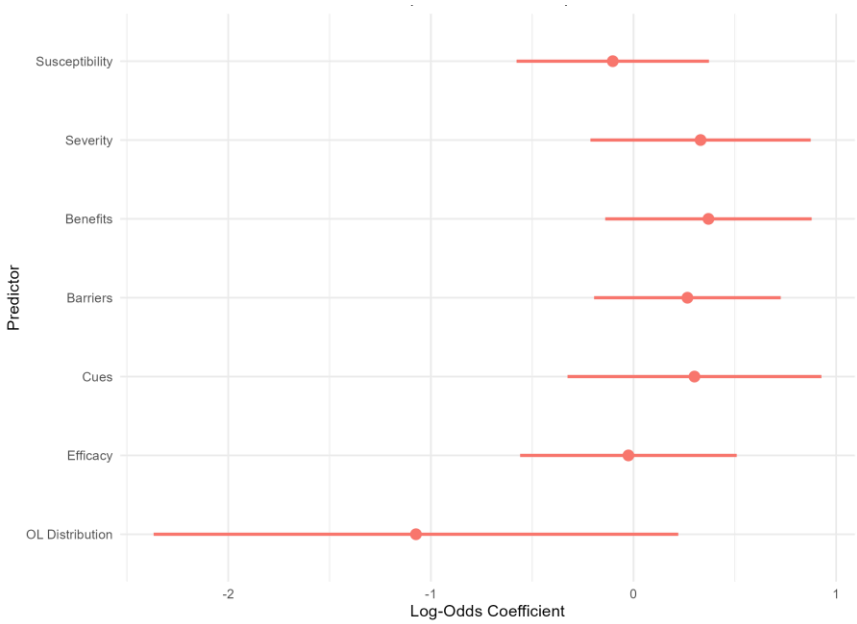


Predictors of Better Understanding of Dengue EWARS

This subsection will discuss the results of the logistic regression done, which looked into the accurate understanding of OL traps as dengue EWARS in the community using the HBM constructs and the presence of the trap in the home.

Across all six HBM constructs, the p-values were too large to be considered statistically significant. This indicates that the earlier finding of the OL trap's success in altering perceived susceptibility to dengue is independent of a person's actual knowledge of the intervention. The regression model shows that high susceptibility has a very slight and non-significant negative relationship with knowledge. This means that even if the person feels susceptible to dengue, it does not necessarily guarantee a functional understanding of EWARS and its role in dengue prevention.

Figure 10. Comparison of Self-Efficacy by OL Distribution Group



Surprisingly, participants in the treatment group had 66% lower odds of demonstrating a correct understanding of the OL trap mechanism (OR = 0.34). This effect, however, is not statistically significant ($p = 0.104$). This finding implies that the receipt of the trap created more confusion than clarity, perhaps due to the lack of proper risk communication measures by the implementing body. Participants who did not receive the trap, relying on existing knowledge and indigenous community beliefs, performed marginally better. This is another critical finding that will be supported by the thematic analysis in the next section of this paper.

Table 2. Coefficients Table of Predictors

	Estimate	**Std. error**	**z value**	**Pr(>|z|)**
(Intercept)	-2.74752	1.77373	-1.549	0.121
Susceptibility	-0.10241	0.24219	-0.423	0.672
Severity	0.33091	0.27734	1.193	0.233
Benefits	0.36979	0.25991	1.423	0.155
Barriers	0.26605	0.23497	1.132	0.258
Cues to action	0.30078	0.31959	0.941	0.347
Efficacy	-0.02511	0.27270	-0.092	0.927
Presence of OL trap	-1.07366	0.66042	-1.626	0.104

This next section of the paper discusses the themes determined and developed by the researchers from the semi-structured interviews done with the residents of Brgy. UP Campus. The following thematic analysis will now be laid out to provide a more nuanced understanding of the two previous quantitative sections.

Theme 1: Agentic Cleaning

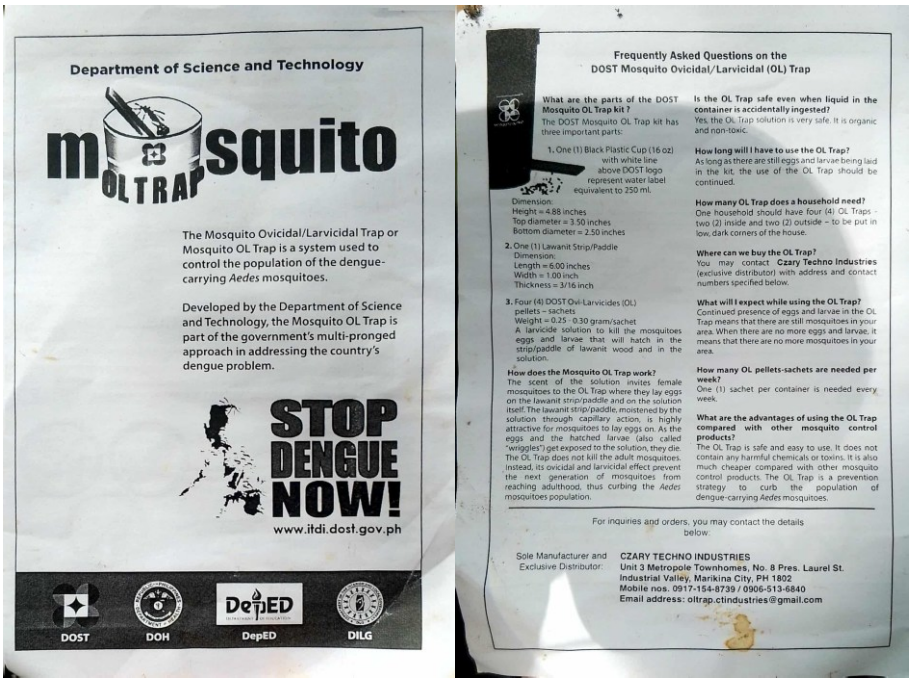
An evident theme that emerged from the participants' responses is the belief that dengue prevention is primarily achieved through general cleanliness, reflected in their frequent response, "basta maglinis ka." This response encapsulates how the residents see their role in mitigating dengue risk: keeping their surroundings clean, regardless of whether they received an OL trap or not, as also supported by the findings presented in Figure 8. The findings reveal that the OL trap, as an intervention and early warning system, does not contribute to how the residents act upon the risk of dengue, but mobilize more from the idea that dengue is more apparent in unsanitary places like canals, and anywhere where stagnant water could linger, like wheels and flower vases.

Theme 2: Lapses in dengue risk communication

The proponents found that there were lapses in terms of risk communication during the rollout of OL traps in Barangay UP Campus. For instance, several respondents have raised questions about the purpose of OL traps, particularly what happens to the samples upon collection. This suggests that recipients of OL traps, albeit interested in its function, may have become passive agents during program implementation, rather than being involved in the process, given their role as primary beneficiaries of its distribution. Hence, this finding exemplifies the lapse in the risk communication initiatives of LGUs to actively involve and engage communities in the process of program implementation, which could have improved the public perception of dengue as a public health issue and induced the constituents to take proactive measures.

Several OL trap recipients have also received brochures from the local government outlining its purpose, as well as the instructions on how to prepare and replace the solution. However, these informational materials, developed by the Department of Science and Technology (DOST), are mainly written in English, particularly the Frequently Asked Questions (FAQs) section of the brochure.

Figure 11. Pages from the OL trap informational brochure



Department of Science and Technology

mosquito OL TRAP

The Mosquito Ovicidal/Larvicidal Trap or Mosquito OL Trap is a system used to control the population of the dengue-carrying *Aedes* mosquitoes.

Developed by the Department of Science and Technology, the Mosquito OL Trap is part of the government's multi-pronged approach in addressing the country's dengue problem.

STOP DENGUE NOW!

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DOST DOH DepED DILG

Frequently Asked Questions on the DOST Mosquito Ovicidal/Larvicidal (OL) Trap

What are the parts of the DOST Mosquito OL Trap kit?
The DOST Mosquito OL Trap kit has three important parts:

1. One (1) Black Plastic Cup (16 oz) with white line above DOST logo represent water label equivalent to 250 ml.
   Dimension:
   Height = 4.88 inches
   Top diameter = 3.50 inches
   Bottom diameter = 2.50 inches
2. One (1) Lawanit Strip/Paddle
   Dimension:
   Length = 6.00 inches
   Width = 1.00 inch
   Thickness = 3/16 inch
3. Four (4) DOST Ovi-Larvicides (OL) pellets – sachets
   Weight = 0.25 - 0.30 gram/sachet
   A larvicide solution to kill the mosquitoes eggs and larvae that will hatch in the strip/paddle of lawanit wood and in the solution.

How does the Mosquito OL Trap work?
The scent of the solution invites female mosquitoes to the OL Trap where they lay eggs on the lawanit strip/paddle and on the solution itself. The lawanit strip/paddle, moistened by the solution through capillary action, is highly attractive for mosquitoes to lay eggs on. As the eggs and the hatched larvae (also called "wriggles") get exposed to the solution, they die. The OL Trap does not kill the adult mosquitoes. Instead, its ovicidal and larvicidal effect prevent the next generation of mosquitoes from reaching adulthood, thus curbing the *Aedes* mosquitoes population.

Is the OL Trap safe even when liquid in the container is accidentally ingested?
Yes, the OL Trap solution is very safe. It is organic and non-toxic.

How long will I have to use the OL Trap?
As long as there are still eggs and larvae being laid in the kit, the use of the OL Trap should be continued.

How many OL Trap does a household need?
One household should have four (4) OL Traps - two (2) inside and two (2) outside – to be put in low, dark corners of the house.

Where can we buy the OL Trap?
You may contact **Czary Techno Industries** (exclusive distributor) with address and contact numbers specified below.

What will I expect while using the OL Trap?
Continued presence of eggs and larvae in the OL Trap means that there are still mosquitoes in your area. When there are no more eggs and larvae, it means that there are no more mosquitoes in your area.

How many OL pellets-sachets are needed per week?
One (1) sachet per container is needed every week.

What are the advantages of using the OL Trap compared with other mosquito control products?
The OL Trap is safe and easy to use. It does not contain any harmful chemicals or toxins. It is also much cheaper compared with other mosquito control products. The OL Trap is a prevention strategy to curb the population of dengue-carrying *Aedes* mosquitoes.

For inquiries and orders, you may contact the details below.

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This explains the prior findings of the proponents that are indicative of the insufficient understanding of households on the relevance of OL traps as an early warning and response system against dengue. Further, this also exhibits the ineffectiveness of risk communication strategies of program implementers, as language barriers and recipients' lack of familiarity with technical terms have not been taken into consideration, thus hindering the improvement of their health literacy.

Figure 12. DEWARS tarpaulin posted in Brgy. UP Campus

CONCLUSION AND RECOMMENDATIONS

Conclusion

This paper comes to three main conclusions, each referring to one of the research questions posed above. First, the paper notes that only susceptibility is increased when intervention is applied in the sample. This means that having an OL trap in the home makes people feel more at risk, a desirable effect to motivate residents of Brgy. UP Campus to take more preventive measures and behaviors. However, the program does not influence confidence, perceived severity of dengue, program benefits, or perception of barriers.

Next, no construct of the HBM, used as a predictor of knowledge, motivates respondents to seek accurate information regarding dengue risks. This, evidently, is a concerning finding, as accurate information regarding dengue is integral in combating the disease proliferation. We note that while the OL trap makes people feel susceptible to dengue, this does not really shape their motivation to gain information regarding it.

Finally, the themes presented herein show the critical lack of community engagement between program implementers and residents. The notable erroneous understanding of residents in how the OL trap functions is a marker of ineffective risk communication. Similarly, this paper noted the technicality and language barriers present in the learning materials for OL traps and dengue.

Overall, this paper concludes that while the OL trap helps shape perceptions of dengue risks, the ineffectiveness of the program, in toto, to help residents understand dengue EWARS through risk communication fails to foster an atmosphere of knowledge that, in turn, could shape tangible preventive behaviors.

Recommendations

Given the findings of this study, the proponents would like to outline the following recommendations for the local government unit of Quezon City, specifically to the Barangay U.P. Campus. Primarily, the local government and in partnership with the concerned national government agencies, such as DOST, DOH, and the UP Resilience Institute, expands the distribution network of OL traps. This is in support of the findings of this study that perceived susceptibility of dengue increases when intervention is applied.

To engage community members on the risks of dengue and the opportunity of EWARS using OL traps, concerned government agencies should implement various methods to inform the public, relative to the literacy level of their constituents. Such information campaigns should be qualified as accessible and understandable for the general public, with consideration to language barriers and varying lifestyle contexts.

In totality, for dengue EWARS to serve its purpose in public health management and risk communication, there is a prevalent need to recalibrate the Epidemic Prone Disease Surveillance Information System (EPDSIS) of the DOH. The revitalization of the Dengue Vector Surveillance System of the DOH also presents an opportunity for appropriate dengue monitoring among communities. The devolved function of public health management among LGUs should also prompt proper resource allocation for the expanded rollout of OL traps, dengue information campaigns, and capacity building programs for Barangay Health Workers (BHWs) in relation to proper data collection, processing, and reporting of crucial community health information.

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